STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45
Today: Sections 5.3 and 7.1.
Last Wednesday: Sections 5.1 and 5.2.

Section 5.1: Joint Probability Distributions

For discrete random variables X and Y, the joint probability mass function (pmf) is

$$p(x,y) = P\big(\{X=x\} \cap \{Y=y\}\big), \qquad \sum_x \sum_y p(x,y) = 1.$$

The marginal pmfs are

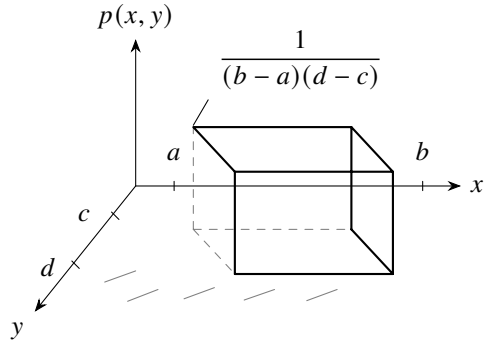
$$p_X(x) = P(X=x) = \sum_{y=-\infty}^{\infty} p(x,y),$$

$$p_Y(y) = P(Y=y) = \sum_{x=-\infty}^{\infty} p(x,y).$$

Example

	$X=0$	$X=1$	$X=2$	
$Y=0$	•	•	•	$p_Y(0)$
$Y=1$	•	•	•	
$Y=2$	•	•	•	
	$p_X(0)$	$p_X(1)$	$p_X(2)$	

Example: $(X,Y) \sim U(a,b,c,d)$



X = calcium level.

Y = cholesterol level.

Model:

$$X \sim U(8.5, 10.5),$$

$$Y \sim U(120, 240).$$

X and Y are independent.

Independence

X and Y are independent if

$$p(x, y) = p_X(x)\, p_Y(y) \qquad \forall (x, y).$$

A failure of factorization such as $p(0, 0) \neq p_X(0)\, p_Y(0)$ rules out independence.

Section 5.2: Expected Values

For discrete X and Y,

$$E[h(X, Y)] = \sum_x \sum_y h(x, y)\, p(x, y),$$

where $p(x, y)$ is the joint pmf.

For continuous X and Y,

$$E[h(X, Y)] = \iint h(x, y)\, p(x, y)\, dx\, dy,$$

where $p(x, y)$ is the joint pdf.

In each case, $E[h(X, Y)]$ is the expected, or mean, value of $h(X, Y)$.

Covariance and Independence

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

If X and Y are independent, then they are uncorrelated. The converse is not necessarily true (see the example in the textbook).

Units of Measurement

Covariance changes when the units of measurement change. If both coordinates are converted from centimeters to inches,

$$\mathrm{Cov}(X, Y) = 0.37 \qquad \text{(measurements in cm)},$$

$$\mathrm{Cov}\left(\frac{X}{2.54}, \frac{Y}{2.54}\right) = \frac{0.37}{2.54^2} \approx 0.0574 \qquad \text{(measurements in inches)}.$$

Section 5.3: Correlation Coefficient

For finite, nonzero standard deviations,

$$\rho \text{ (rho)} = \mathrm{Corr}(X, Y) = \frac{\overbrace{\mathrm{Cov}(X, Y)}^{\text{covariance}}}{\mathrm{sd}[X]\ \mathrm{sd}[Y]}.$$

Properties

1. $-1 \leq \rho \leq 1$.
2. Changing units of measurement by positive conversion factors does not change ρ.
3. If X and Y are independent, then $\rho = 0$; X and Y are uncorrelated.
4. If $Y = aX + b$ (a, b constants), then $\rho = +1$ for $a > 0$ and $\rho = -1$ for $a < 0$.

Proof of Property 4

Let $Y = aX + b$. Then

$$E[Y] = aE[X] + b.$$

$$\begin{aligned} V[Y] &= E\left[(Y - E[Y])^2\right] \\ &= E\left[(aX + b - (aE[X] + b))^2\right] \\ &= E\left[(aX - aE[X])^2\right] \\ &= a^2 E\left[(X - E[X])^2\right] \\ &= a^2 V[X]. \end{aligned}$$

So

$$\begin{aligned} \text{sd}[Y] &= \sqrt{V[Y]} = \sqrt{a^2 V[X]} \\ &= \sqrt{a^2}\,\sqrt{V[X]} \\ &= |a|\ \text{sd}[X]. \end{aligned} \tag{3}$$

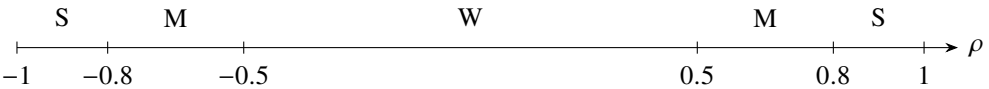
Also,

$$\begin{aligned} \text{Cov}(X, Y) &= E\left[(X - E[X])(Y - E[Y])\right] \\ &= E\left[(X - E[X])(aX + b - (aE[X] + b))\right] \\ &= E\left[(X - E[X])\, a\, (X - E[X])\right] \\ &= aE\left[(X - E[X])^2\right] \\ &= aV[X]. \end{aligned} \tag{4}$$

Therefore,

$$\begin{aligned} \rho &= \frac{\text{Cov}(X, Y)}{\text{sd}[X]\ \text{sd}[Y]} \\ &= \frac{aV[X]}{\text{sd}[X]\,|a|\ \text{sd}[X]} \qquad \text{by (3) and (4)} \\ &= \frac{aV[X]}{|a|V[X]} = \frac{a}{|a|} = \begin{cases} +1, & a > 0, \\ -1, & a < 0. \end{cases} \end{aligned}$$

Strength of Correlation



S: strong M: moderate W: weak

In Practice: Bivariate Data

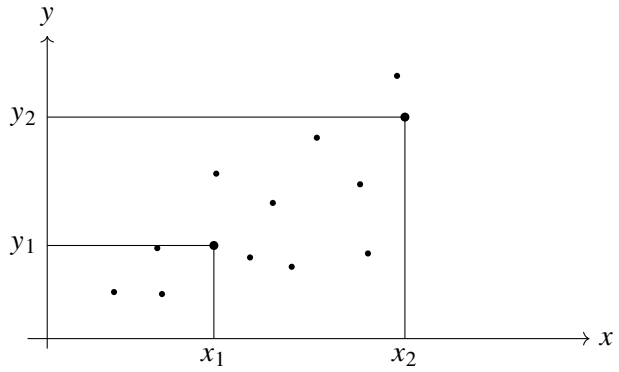
$$X = \text{independent variable}, \qquad Y = \text{dependent variable}.$$

A bivariate dataset is

$$\big\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\big\}.$$

Bivariate data are part of multivariate statistics.

Scatterplot



Sample Correlation Coefficient

The correlation coefficient ρ is estimated by the sample correlation coefficient R. For $n > 1$ and $S_x, S_y > 0$,

$$R = \frac{1}{n-1} \sum_{i=1}^{n} z_{x_i}\, z_{y_i}.$$

The z-scores are

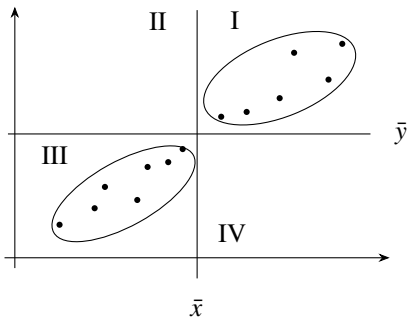
$$z_{x_i} = \frac{x_i - \bar{x}}{S_x}, \qquad z_{y_i} = \frac{y_i - \bar{y}}{S_y},$$

where

$$\bar{x} = \text{sample mean (average)} = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

$$S_x = \text{sample standard deviation} = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}.$$

Case I



In quadrant I,

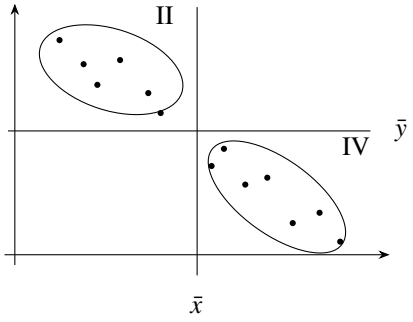
$$\left.\begin{aligned} z_{x_i} &> 0 \\ z_{y_i} &> 0 \end{aligned}\right\} \quad z_{x_i} z_{y_i} > 0.$$

In quadrant III,

$$\left.\begin{aligned} z_{x_i} &< 0 \\ z_{y_i} &< 0 \end{aligned}\right\} \quad z_{x_i} z_{y_i} > 0.$$

Large positive value of R.

Case II



Quadrant II: $z_{x_i} z_{y_i} < 0$,

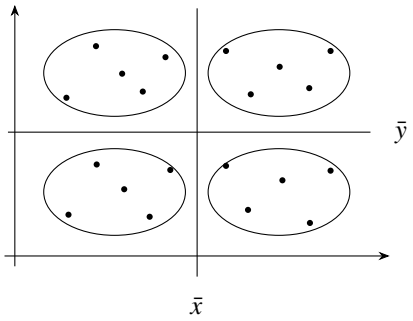
Quadrant IV: $z_{x_i} z_{y_i} < 0$.

Large negative value of R.

$$R = -0.87.$$

Strong negative correlation.

Case III



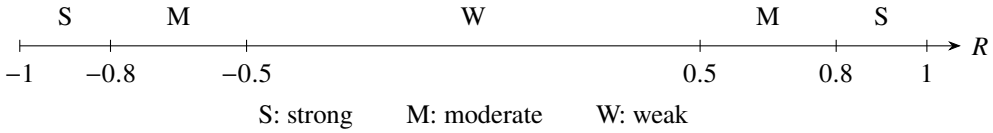
The products

$$z_{x_i} z_{y_i} < 0 \quad \text{and} \quad z_{x_i} z_{y_i} > 0$$

cancel each other out.

$$R \approx 0.$$

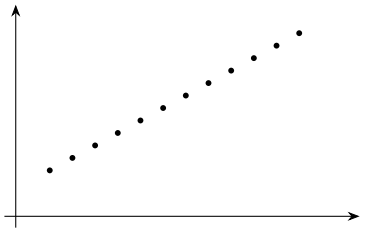
R is a measure of the linear dependence between x and y.



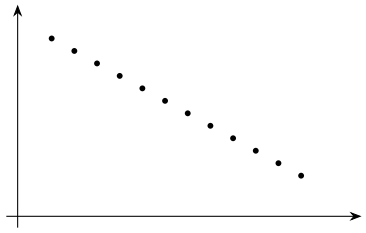
Another Formula for R

$$R = \frac{1}{n-1}\sum_{i=1}^{n} z_{x_i} z_{y_i} = \frac{\displaystyle\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\left[\displaystyle\sum_{i=1}^{n}(x_i - \bar{x})^2\right]\left[\displaystyle\sum_{i=1}^{n}(y_i - \bar{y})^2\right]}}.$$

If all observations satisfy $y = a + bx$, with nonconstant x-values and $b \neq 0$, then $R = \pm 1$.



$R = +1$



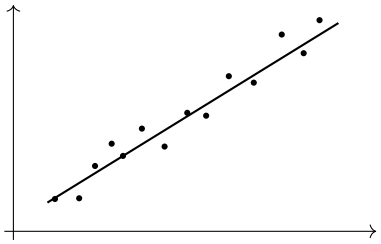
$R = -1$

Estimating ρ

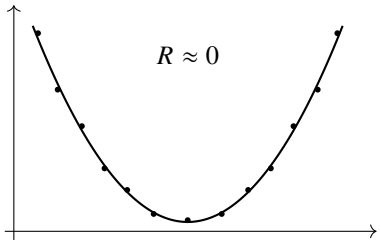
$$\rho = ? \qquad \widehat{\rho} = R.$$

An estimator for ρ is given by R.

Linear and Nonlinear Regression



Linear regression
Chapter 11



Nonlinear regression

The Line of Best Fit

The line of best fit is also called the linear regression line or the least-squares line. Its equation is

$$y = a + bx.$$

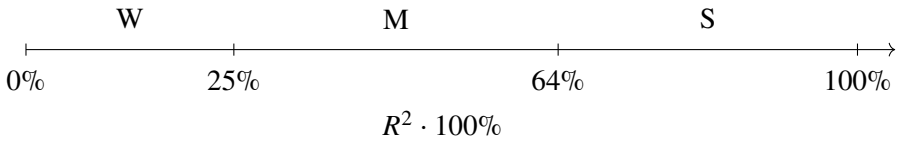
$$b = R\,\frac{S_y}{S_x} \qquad \text{slope,}$$

$$a = \bar{y} - b\bar{x} \qquad y\text{-intercept.}$$

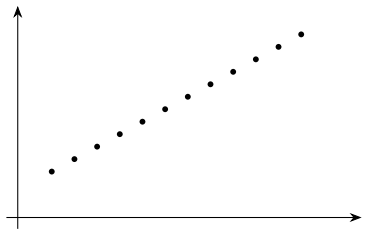
Coefficient of Determination

$$R^2 = \text{coefficient of determination.}$$

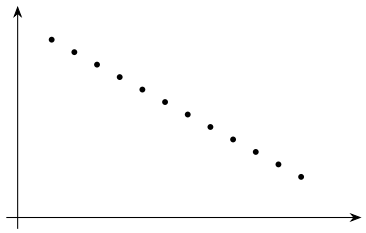
$R^2 \cdot 100\%$ is the percentage of the total variation in the y-observations that can be explained by the linear dependence on x.



$R^2 \cdot 100\%$



$R^2 \cdot 100\% = 100\%$



$R^2 \cdot 100\% = 100\%$

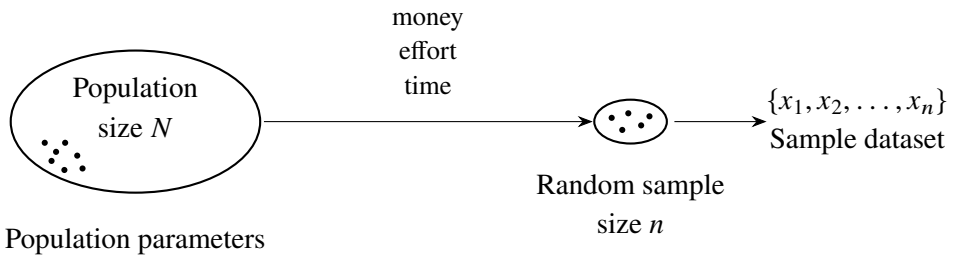
Homework

Chapter 5: 30, 32, 36, 38, 55, 56.

Chapter 6: 12, 14, 20, 24, 26, 31.

Chapter 7: Estimation

Section 7.1: Point Estimation



Here x_i is the ith measurement, or observation. The sample dataset gives sample statistics.

Population Parameters and Sample Statistics

Population parameter (unknown)		Sample statistic (known)
μ: population mean (average)	$\longrightarrow$	$\bar{x}$: sample average
σ: population standard deviation	$\longrightarrow$	S: sample standard deviation
ρ: population correlation coefficient	$\longrightarrow$	R: sample correlation coefficient

Point Estimator and Point Estimate

A population parameter is typically unknown. If a sample statistic is used to estimate a population parameter, then the sample statistic is called a point estimator. Its actual value for the sample dataset is called the point estimate.

If θ (theta) denotes a population parameter, then $\widehat{\theta}$ denotes the point estimator. Greek letters denote population parameters; a hat indicates an estimator.

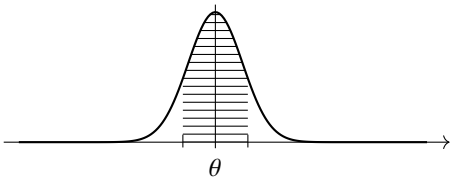
Examples

$$\mu = ? \qquad \widehat{\mu} = \underbrace{\bar{x}}_{\text{point estimator}} = \underbrace{3.17}_{\text{point estimate}} .$$

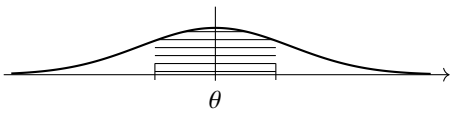
$$\widehat{\sigma} = S \quad \text{(sample standard deviation)}, \qquad \widehat{\rho} = R.$$

Desirable Properties of a Point Estimator

1. Small bias: $E[\widehat{\theta}] - \theta$ is close to zero.
2. Small standard deviation: $\text{sd}(\widehat{\theta})$.



Sampling distribution of $\widehat{\theta}_1$



Sampling distribution of $\widehat{\theta}_2$

$$E[\widehat{\theta}_1] = E[\widehat{\theta}_2], \qquad \text{sd}(\widehat{\theta}_1) < \text{sd}(\widehat{\theta}_2).$$